

Electric Motor Maintenance Manual

Safety

Disconnect and lock out electrical power before performing any inspection or maintenance on the motor. Verify the circuit is de-energized with a calibrated meter before touching any terminal. Allow the motor housing to cool before handling; surface temperatures can exceed 70C during normal operation. Do not operate the motor with guards or covers removed.

Installation

Mount the motor on a rigid, level base to minimize vibration transfer. Align the motor shaft with the driven load within the coupling manufacturer's tolerance before first startup. Confirm supply voltage and frequency match the motor nameplate rating. Torque all foot-mounting bolts to the value specified in the installation drawing.

Troubleshooting

Excessive vibration

Excessive vibration is most commonly caused by shaft misalignment, an unbalanced coupling, a loose foot mounting, or worn bearings. Check alignment first, since it accounts for the majority of vibration complaints in the field. If vibration persists after realignment, inspect the coupling for wear and the bearings for play. Vibration RMS values above 4.5 mm/s at the bearing housing generally warrant further investigation.

Overheating

Overheating is typically caused by sustained overload, blocked ventilation passages, low supply voltage, or a failing bearing increasing friction. Confirm the load current does not exceed the nameplate full-load amperage. Clear any dust or debris from the cooling fins and fan shroud. A winding temperature more than 40C above ambient during normal load is a signal to schedule inspection before the insulation degrades further.

Unusual noise

A grinding or rumbling noise usually points to bearing damage or contamination inside the bearing race. A high-pitched whine can indicate the motor is running at the wrong voltage or

frequency. A rhythmic knocking that varies with shaft speed is often a bent shaft or a loose rotor component. Isolate the noise source before disassembly by listening near each bearing housing independently.

Maintenance

Bearings

Inspect bearings for excessive play, noise, or heat at the interval specified in the maintenance schedule for the duty class. Replace bearings in pairs rather than individually to avoid re-introducing imbalance shortly after service. Use only the bearing part number specified for the motor frame size.

Lubrication

Re-grease sealed bearings only if the motor is designed for re-lubrication; over-greasing is a common cause of premature bearing failure. Use the grease type specified on the motor nameplate — mixing incompatible greases can cause the lubricant to break down. Wipe away any purged grease after re-lubrication.